

SIMATS ENGINEERING

CO4-AT1-INDUSTRIAL PROBLEM SOLVING

COURSE CODE: CSA0904

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COURSE NAME: Programming in JAVA

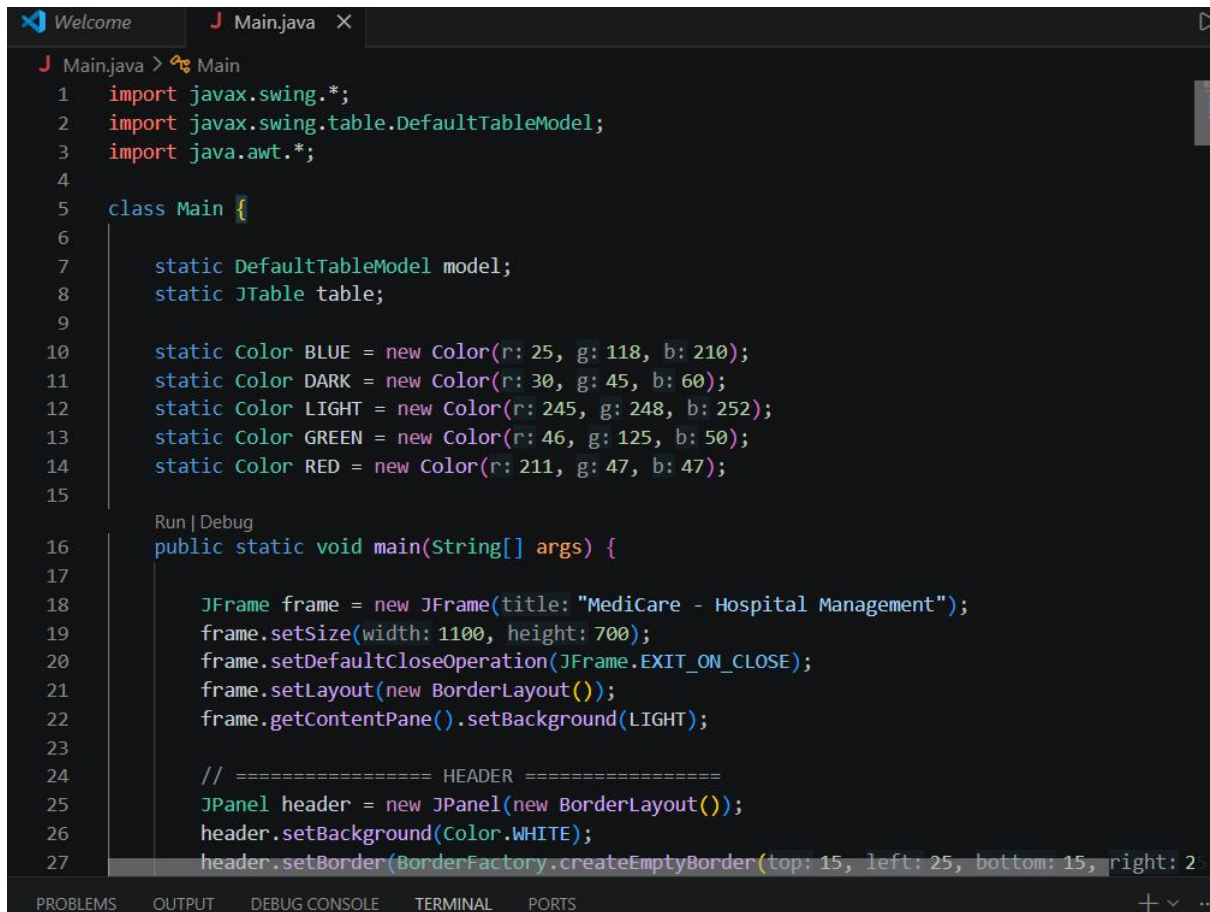
REG NO: 192421410

Q1 Hospital Appointment Management System

50 marks

Design a Swing-based GUI application for a hospital to manage patient appointments, including booking, updating, and canceling schedules. Use AWT/Swing controls such as buttons, text fields, tables, and menus to facilitate user interaction. Implement event listeners to handle actions like appointment confirmation and deletion. Ensure the system can handle dynamic patient data efficiently. Evaluate different layout managers (FlowLayout, GridLayout, BorderLayout) for organizing complex UI components. Analyze how the design supports scalability when the number of patients increases. Propose improvements for better usability and real-time updates.

PROGRAM:



```
1  import javax.swing.*;
2  import javax.swing.table.DefaultTableModel;
3  import java.awt.*;
4
5  class Main {
6
7      static DefaultTableModel model;
8      static JTable table;
9
10     static Color BLUE = new Color(r: 25, g: 118, b: 210);
11     static Color DARK = new Color(r: 30, g: 45, b: 60);
12     static Color LIGHT = new Color(r: 245, g: 248, b: 252);
13     static Color GREEN = new Color(r: 46, g: 125, b: 50);
14     static Color RED = new Color(r: 211, g: 47, b: 47);
15
16     public static void main(String[] args) {
17
18         JFrame frame = new JFrame(title: "MediCare - Hospital Management");
19         frame.setSize(width: 1100, height: 700);
20         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
21         frame.setLayout(new BorderLayout());
22         frame.getContentPane().setBackground(LIGHT);
23
24         // ===== HEADER =====
25         JPanel header = new JPanel(new BorderLayout());
26         header.setBackground(Color.WHITE);
27         header.setBorder(BorderFactory.createEmptyBorder(top: 15, left: 25, bottom: 15, right: 25));
```

OUTPUT:

Hospital Appointment Management

Menu

HOSPITAL APPOINTMENT SYSTEM

Patient Details

Patient Name:
Vikram

Age:
38

Doctor:
Dr. Ravi - Dental

Date:
30.8.26

Appointment Time

2:00 pm

BOOK
UPDATE
CANCEL
CLEAR

Appointments

Patient Name	Age	Doctor	Date	Time
Divya	24	Dr. Priya - Cardiology	20.8.26	6:00 pm
Vikram	38	Dr. Ravi - Dental	30.8.26	2:00 pm

Success

Appointment Booked Successfully!

OK

Q2 Online Banking Interface

50 marks

Develop a GUI-based banking application that allows users to perform operations such as fund transfer, balance inquiry, and viewing transaction history. Use Swing components and event listeners to manage user interactions securely. Implement input validation for sensitive data like account number and amount. Ensure proper error handling for invalid transactions. Apply suitable layout managers to create a responsive and user-friendly interface. Analyze system responsiveness during multiple operations and suggest improvements. Demonstrate how secure input handling enhances reliability.

PROGRAM:

```

Welcome Main.java
Main.java > Main > BLUE
1  import javax.swing.*;
2  import javax.swing.table.DefaultTableModel;
3  import java.awt.*;
4
5  class Main {
6
7      static double balance = 50000;
8      static DefaultTableModel model;
9
10     static Color NAVY = new Color(r: 20, g: 35, b: 55);
11     static Color BLUE = new Color(r: 45, g: 100, b: 220);
12     static Color BG = new Color(r: 242, g: 245, b: 250);
13     static Color GREEN = new Color(r: 35, g: 150, b: 90);
14     static Color RED = new Color(r: 210, g: 60, b: 60);
15
16     public static void main(String[] args) {
17
18         JFrame f = new JFrame(title: "NovaBank Online Banking");
19         f.setSize(width: 1100, height: 700);
20         f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
21         f.setLayout(new BorderLayout());
22         f.getContentPane().setBackground(BG);
23
24         // ===== TOP NAVIGATION =====
25
26         JPanel nav = new JPanel(new BorderLayout());
27         nav.setBackground(NAVY);

```

OUTPUT:

